What robot pretraining buys a vision-language-action model

A controlled component-interaction study across two embodiments

`Six arms · one frozen-backbone head · 2,600 closed-loop rollouts · LIBERO-Goal and ALOHA transfer-cube`

Existing work studies how to choose data and how to choose a backbone. This study asks what happens *between* the components: how backbone pretraining, camera configuration, action space and task structure interact to decide whether a policy actually succeeds in the loop.

The six claims

#	Claim	Key number
C1	A camera matters far more than a pretrained backbone	+29.0 vs +2.5 pts
C2	Pretraining helps on one task and not the other, at every stage of training	+9.5 pts, p = 0.0067
C3	The whole advantage sits in one transition	p = 0.0009
C4	The gap is in failure rate, not kind — so offline metrics miss it	<2% vs 15.5%
C5	Language ablation has a floor that must be measured	1.14–1.24×
C6	Visual reliance tracks what the action is defined relative to	0.038 vs 0.11

Setup

One identical 19.2M-parameter head — token adapter plus DiT flow-matching decoder — is trained against **frozen** published backbones and compared by closed-loop rollout. Every arm shares the head, the read depth, the optimiser schedule and the evaluation protocol, so a difference in success rate is attributable to the factor that moved.

The pair spans one lineage: stock **Qwen3-VL-2B** → Cosmos-Reason2-2B → **GR00T N1.7**. Both hops are verified real finetunes (584/625 and 476/493 tensors differ), so the comparison covers the whole robot-pretraining treatment rather than one weak hop.

Two controls make the pair comparable. **Layer matching:** both arms are read at layer 16 — GR00T's own `select_layer`, intermediate for Qwen3-VL's 28 layers — because reading each at its own last layer would compare depth 16 against depth 28 and attribute a depth effect to pretraining. **Final-norm correction:** since layer 16 is final for one arm and intermediate for the other, the language stack's final RMSNorm is applied to the intermediate read, without which the pair would differ by normalisation rather than by weights.

C1 · A camera matters far more than a pretrained backbone

CLAIM 1

On single-arm manipulation, observation configuration outweighs backbone pretraining by an order of magnitude.

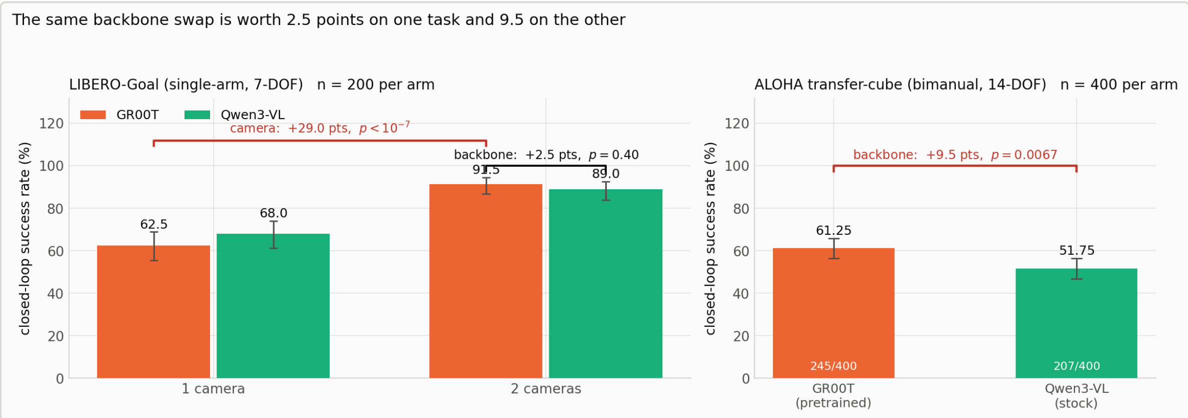
EVIDENCE

Arm	Backbone	Cameras	Success (n = 200)
exp03	GR00T N1.7	1	62.5%
exp04	Qwen3-VL-2B	1	68.0%
exp05	GR00T N1.7	2	**91.5%**
exp06	Qwen3-VL-2B	2	**89.0%**

Adding the wrist camera: **+29.0 points** (GR00T) and **+21.0 points** (Qwen3-VL), $p < 10^{-7}$. Swapping the backbone at the benchmark's own two-camera specification: **+2.5 points**, p = 0.40.

Supporting detail: the instruction is load-bearing for every arm (0/200 under a swapped instruction, all ten tasks), and both two-view arms converge to the same routing despite different backbone weights — wrist attention 51.6% vs 54.5%, ablation cost ×11.94 vs ×11.64.

FIGURE



The study in one figure. Left: on LIBERO-Goal a second camera is worth +29.0 points while swapping the backbone at that same observation spec is worth 2.5 (p = 0.40). Right: on bimanual ALOHA the identical backbone swap is worth +9.5 points (p = 0.0067). Error bars are Wilson 95% intervals.

Bounded to: one task suite (LIBERO-Goal) and one backbone pair. The single-view arms are reported for completeness but carry no claim — single-view puts the pretrained arm outside its trained input configuration while leaving the stock arm inside its own, an asymmetry pointing the same way as the result.

C2 · Pretraining's payoff is task-specific, at every stage of training

CLAIM 2

Robot pretraining is not a general sample-efficiency prior: it is decisive on one task and absent on the other, from the first checkpoint to the last.

EVIDENCE — THE BIMANUAL TASK

ALOHA transfer-cube was chosen as the pre-registered falsifier for C1's null. Bimanual manipulation is inside GR00T's pretraining distribution, so the venue is biased *toward* finding an effect; a null here would have been strong evidence. It did not return a null.

Arm	Run 1	Run 2	Pooled (n = 400)	Wilson 95% CI
GR00T N1.7	60.0%	62.5%	**61.25%**	[56.4, 65.9]
Qwen3-VL-2B	49.0%	54.5%	51.75%	[46.9, 56.6]

+9.5 points, z = 2.71, p = 0.0067 — clears the Bonferroni bar for the six comparisons in this study (0.05/6 = 0.0083), and the intervals do not overlap. The two runs used disjoint seed ranges, so this is a genuine replication rather than a resampling of the policy's own noise.

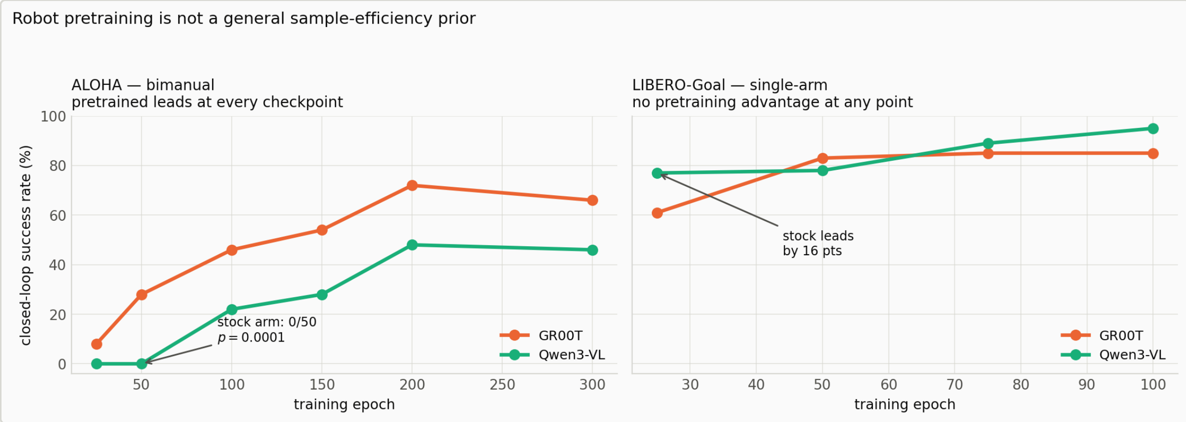
EVIDENCE — THE WHOLE TRAINING CURVE

ALOHA epoch	25	50	100	150	200	300
GR00T	8.0%	28.0%	46.0%	54.0%	72.0%	66.0%
Qwen3-VL	**0.0%**	**0.0%**	22.0%	28.0%	48.0%	46.0%

LIBERO epoch	25	50	75	100
GR00T	61.0%	83.0%	85.0%	85.0%
Qwen3-VL	**77.0%**	78.0%	89.0%	**95.0%**

> On ALOHA the stock backbone **cannot do the task at all** before epoch 100 — 0/50 at epochs 25 and 50 while the pretrained arm is at 28% (McNemar p = 0.0001) — and needs ~2× the epochs to reach any given rate. On LIBERO the *stock* arm leads at epoch 25 by 16 points. The early-training effect, the largest in the study, does not transfer.

FIGURE



Both checkpoint ladders on a common axis. Left: on bimanual ALOHA the pretrained arm leads at every checkpoint. Right: on single-arm LIBERO-Goal it leads at none. Paired seeds throughout; LIBERO's 50 fixed initial states make its pairing exact.

Bounded to: the two testbeds differ on *six axes simultaneously* — instruction variation (10 vs 1), degrees of freedom (7 vs 14), arm count, action space, camera count, and distribution match to GR00T's pretraining corpus. The DOF hypothesis, the bimanual hypothesis and the distribution-match hypothesis all predict what was observed, and this design cannot separate them. Naming bimanual coordination as *the* cause is an interpretation, not a result.

Two further cautions: on LIBERO neither nominally significant point ($p = 0.0195$, 0.0213) survives Bonferroni over four comparisons, so the claim there is the negative one. And both ladders disagree with their own headline levels — read the ladders for *shape*, the pooled runs for *magnitude*.

C3 · The whole advantage sits in one transition

CLAIM 3

Pretraining does not raise general competence — it raises one conditional probability.

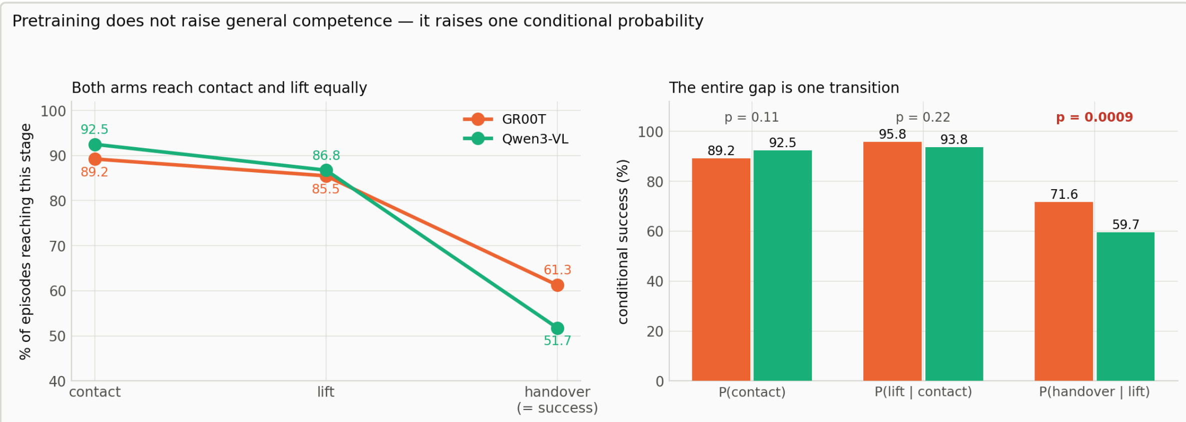
EVIDENCE

ALOHA scores contact / lift / handover / success. `max_reward == 3` never occurs in 800 episodes, so once the receiving gripper touches the cube the episode always completes, and the ladder is contact → lift → handover.

Stage	GR00T	Qwen3-VL	Δ	p
P(contact)	89.2%	92.5%	−3.3	0.11
P(lift ∣ contact)	95.8%	93.8%	+2.0	0.22
P(handover ∣ lift)	**71.6%**	**59.7%**	**+12.0**	**0.0009**

Both early stages run slightly *against* the pretrained arm, and the localisation is tighter ($p = 0.0009$) than the top-line result ($p = 0.0067$). Decomposing success along the environment's own reward ladder converts an aggregate difference into a mechanism at no additional rollout cost — a method other closed-loop studies can adopt directly.

FIGURE



Left: both arms reach contact and lift at the same rate — the curves separate only at the final transition. Right: the same data as conditional probabilities. P(handover ∣ lift) carries the entire gap.

Bounded to: one task. Whether the pattern generalises to other multi-stage manipulation tasks is untested.

C4 · The gap is in failure rate, not failure kind

CLAIM 4

Offline metrics cannot rank these policies — and the instrumented rollouts show exactly why.

EVIDENCE — THE BLINDNESS

Measure (stock – pretrained, % of pretrained)	Δ
velocity loss (overall)	+0.1%
velocity loss (handover phase)	+1.9%
open-loop action error (nMAE, 14 dims)	−0.4%
PE sensitivity	+1.1%
attention mass on image	−5.2%
closed-loop success rate	**−15.5%**

Four accuracy measures differ by under 2% against a gap the rollouts resolve at $p = 0.0067$. The training objective *does* register that the handover window is hard — mid-phase loss is 1.8× the late-phase value for both arms — but not that one arm is worse at it.

EVIDENCE — THE REASON

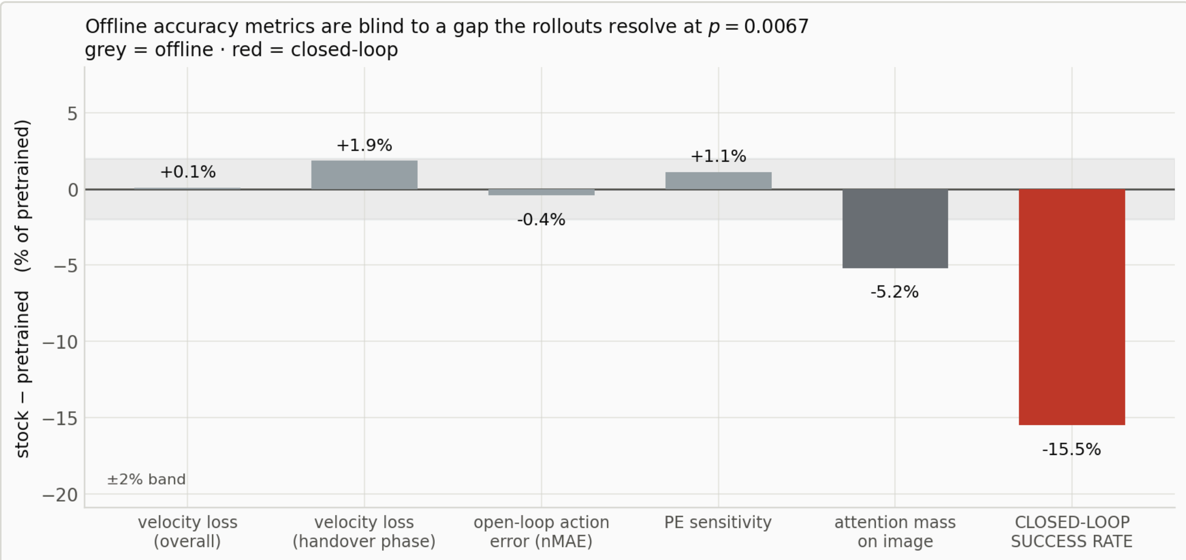
Instrumented rollouts ($n = 200$ per arm) log per-step 14-DOF state, commanded action, the reward timeline and per-replan attention. Arm identity was *measured*, not assumed: the picking arm is whichever block moves first in successful episodes. Classifying episodes that lifted the cube but failed the handover:

Failure class	GR00T	Qwen3-VL	p
premature receiver close	59.6% (31)	59.4% (41)	**0.98**
receiver never engaged	19.2% (10)	26.1% (18)	0.38
grasp lost after lift	21.2% (11)	14.5% (10)	0.34
total lifted-but-failed	**52**	**69**	—

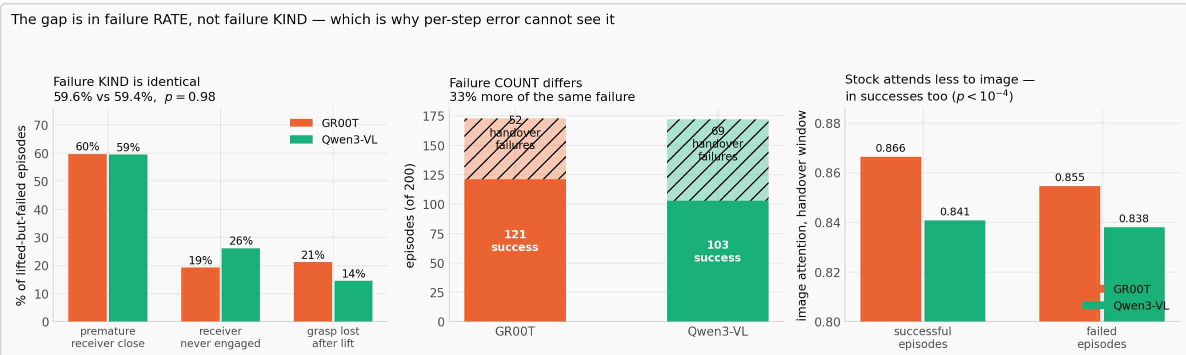
> The failure **mix is statistically identical; only the count differs** — 33% more of the same failure. Two policies that fail the same way, differing only in how often they enter an unrecoverable configuration, leave **no per-step action-error signature**. The difference is a compounding closed-loop property, not a per-step accuracy property — exactly what an error averaged over a 16-step chunk and 2,000 frames cannot see.

Attention was recorded **during rollout**, not on dataset frames: dataset frames are successful expert demonstrations and cannot be linked to the policy's own failures. Over the five replans after the lift the stock arm allocates less mass to image tokens — 0.8407 vs 0.8663 in successful episodes (Welch t = 9.01) and 0.8380 vs 0.8546 in failed ones (t = 5.24), both $p < 10^{-4}$.

FIGURES



Every offline diagnostic as a between-arm difference, on the same axis as the closed-loop result. Four accuracy measures sit inside the ±2% band.



Instrumented rollouts. Left: the failure mix is identical. Centre: only the count differs. Right: image attention during the handover window, split by outcome so the comparison cannot be an artifact of failed episodes running longer.

Bounded to: these are correlational measurements on frozen checkpoints; they locate where two policies differ but cannot prove the difference causes the gap. The absolute class proportions depend on the gripper-closure threshold, though the between-arm comparison — 59.6% vs 59.4% — does not. And because the attention difference appears in successful episodes too, it is a stable policy-level habit, *not* a signature of impending failure; it must not be described as the policy recognising anything.

C5 · Language ablation has a floor that must be measured

CLAIM 5

Zeroing text tokens costs something even when the text carries no information — so ablation ratios must be read against that floor, not against 1.0.

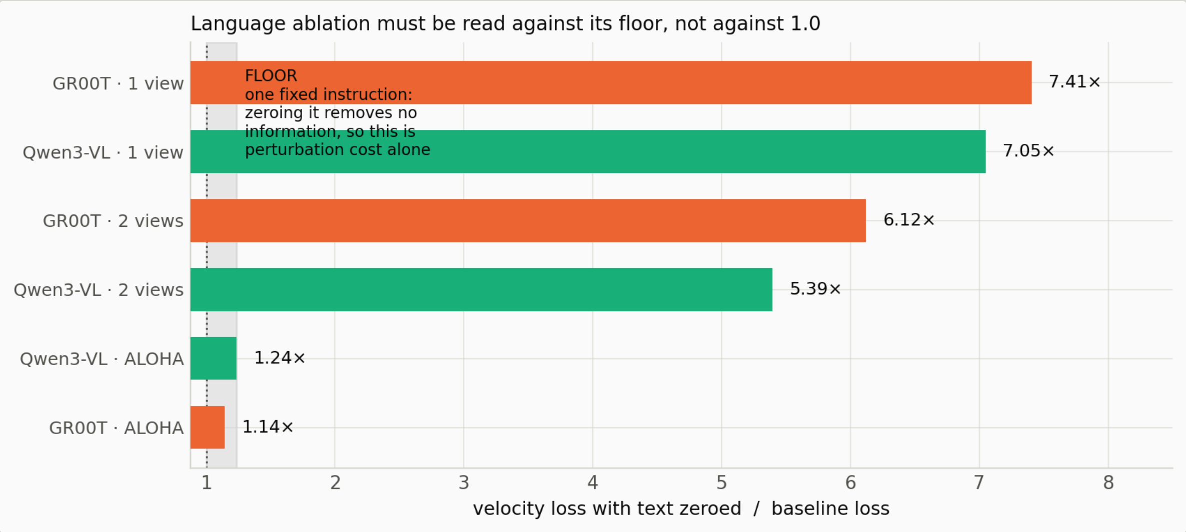
EVIDENCE

ALOHA has a single fixed instruction, so zeroing its text tokens removes no task-discriminative information. Loss nonetheless rises **1.14×** (GR00T) and **1.24×** (Qwen3-VL). That residual is the cost of an off-distribution perturbation alone — a control condition an ablation on a multi-instruction dataset cannot supply for itself.

Arm	Instructions	Ablation ratio
GR00T · ALOHA	1 (fixed)	1.14× — *floor*
Qwen3-VL · ALOHA	1 (fixed)	1.24× — *floor*
Qwen3-VL · 2 views	10	5.39×
GR00T · 2 views	10	6.12×
Qwen3-VL · 1 view	10	7.05×
GR00T · 1 view	10	7.41×

A second reading falls out of the same table: the two-view arms depend on language measurably *less* than the one-view arms (5.39×/6.12× against 7.05×/7.41×), consistent with a wrist camera recovering information the instruction previously had to supply.

FIGURE



Text-ablation ratio across all six arms. The grey band is the floor measured on ALOHA's single fixed instruction — the cost of the perturbation alone.

> **Bounded to:** the floor is measured on one dataset and one head; its magnitude on other setups is unknown. The methodological point — that a floor exists and is not 1.0 — does not depend on its exact value.

C6 · Visual reliance tracks what the action is defined relative to

CLAIM 6

A policy leans on image position when the action is a displacement it cannot infer from one frame, and not when the action is a destination the frame already implies.

EVIDENCE

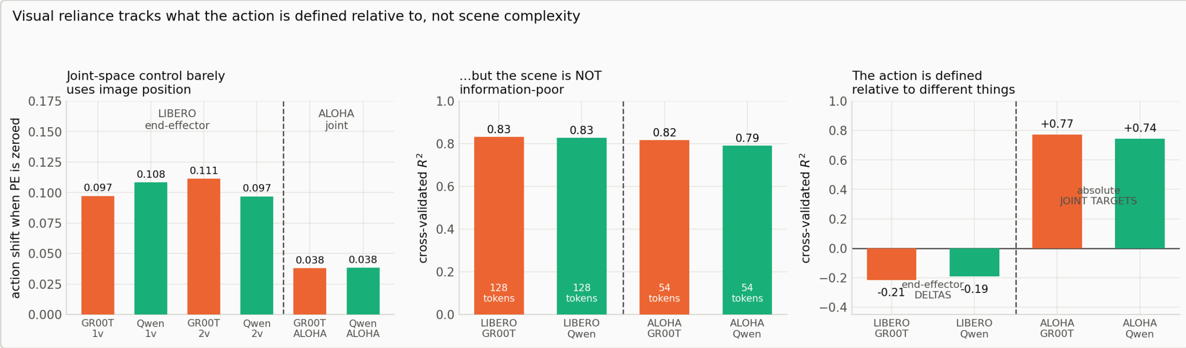
Zeroing the 2D positional encoding shifts actions by **0.097–0.111** under end-effector control but only **0.038** under 14-DOF joint-space control with full proprioception.

The obvious objection is that ALOHA's scene is simply visually simpler. A gradient or Jacobian analysis cannot settle that — an impoverished scene produces low image sensitivity under either explanation. A probe of what the tokens *contain* can:

Arm	Image tokens	R^2(arm state)	R^2(action)
LIBERO GR00T 2-view	128	0.832	−0.215
LIBERO Qwen3-VL 2-view	128	0.828	−0.192
ALOHA GR00T	54	0.817	0.771
ALOHA Qwen3-VL	54	0.791	0.743

> Arm configuration is **equally readable on both testbeds** (0.830 vs 0.804) — and ALOHA reaches that from **54 image tokens against LIBERO's 128**. The scene is not information-poor, so "visually simpler" does not explain the lower PE sensitivity.
>
> The second column identifies the operative variable. LIBERO commands end-effector *deltas*, which one frame does not determine (R^2 −0.20); ALOHA commands *absolute joint targets*, which sit close to the pose the image already encodes (R^2 +0.76). The driver is what the action is defined relative to — not degrees of freedom, not arm count.

FIGURE



Left: PE sensitivity across all six arms. Centre: arm configuration is equally readable from image tokens on both testbeds, and ALOHA does it with fewer tokens. Right: action recoverability from a single frame, the variable that separates them.

Bounded to: the action-recoverability contrast is *partly definitional* — absolute targets are near the current state by construction — so it is reported as *explaining* the PE result rather than as independent evidence for it. Token geometry also differs between the testbeds (54 vs 128) and is not separately controlled.

Validity

The harness is correct. With the policy removed from the loop and the demonstrations' own actions replayed through the same env construction, success detector and step budget, LIBERO replays at 90.0% (no settling) and 92.0% (5 settling steps) — the band published methods report, so this is not a stricter harness. It is a replay rate, not a policy ceiling: the two-view arms exceed it on two tasks.

Training budget is not the constraint. Each arm takes 53,760 optimizer steps at batch 128 — 7.2× more samples and ~34× more passes over the data than a published LeRobot-family recipe on LIBERO. Validation flattens after epoch ~75 and open-loop action correlation reaches 0.979–0.997.

Instrumentation did not perturb the policies. The traced run reproduced 60.5% / 51.5% against the original 60.0% / 49.0% on the same seeds.

What this study cannot settle

- **Two backbones, one lineage.** Every backbone claim rests on one pair.
- **The six-axis confound** (see C2) — two testbeds cannot separate six covarying differences.
- **One head architecture.** "Backbone barely matters on LIBERO" may be specific to a head with this capacity and cross-attention design.
- **Correlational diagnostics.** A causal test would swap handover-window tokens between arms.
- **Simulation only.** No physical robot.
- **Nulls are not equivalence.** At n = 200 per LIBERO condition the standard error is 3.3 points.

Summary

> Robot pretraining of a frozen VLM backbone is **not a general sample-efficiency prior**. Its payoff is task-specific, and where it fails it fails at every point in training rather than only at convergence.
>
> On single-arm pick-and-place at the benchmark's own two-camera specification it contributes nothing measurable — 2.5 points at $p = 0.40$, and no advantage at any checkpoint — while a second camera contributes +21 to +29. On bimanual handover it is the difference between 0% and 28% success at 50 epochs, halves the epochs to any target, and leaves a replicated +9.5-point advantage that localises to a single transition.
>
> Throughout, the offline metrics these systems are trained on and selected by failed to rank the resulting policies — and the instrumented rollouts show why: the policies fail in the same way, differing only in how often.

`Full design, statistics and defect log:` `RESULTS.md` `· Framing and component ledger:` `OBJECTIVE.md` `· Figures:` `scripts/analysis/plots_paper.py`